

1. The raw audio is a long list of amplitude values over time. It's difficult for neural networks to learn from this, as it means that the same word spoken faster, louder, or by a different person would look completely different. By converting the raw audio into a spectrogram or MFCC features, we end up extracting which frequencies are present and when rather than relying on just amplitude values over time, which are way more consistent and meaningful for speech. MFCCs help compress this even further, keeping the input small enough to run on a small microcontroller.
2. When doing full integer quantization, all the model's 32-bit floats get converted into 8-bit integers to make it smaller and faster. However, in order to do so without affecting the accuracy, the quantization process needs to know what range of values each layer actually sees during inference. The representative dataset acts as a small sample of real inputs that can be run through the model, from which we can observe activation ranges and select good scale factors.
3. `audio_processor.get_data()` turns raw audio files into features that the model can actually use. It works by reading the .wav file, running it through a preprocessing pipeline (FFT, spectrogram, MFCC), and returning the processed feature arrays for us to use as training or inference. It can also handle train/test splits and mixing in background noise.
4. `g_model` is the actual TFLite model, which is a C byte array containing the entire TFLite model binary, converted into hex values. `g_model_len` is the length of that array in bytes. The Arduino has no way to load a file at runtime (since it has no filesystem), so the model has to be loaded directly into the code as this bite array.
- 5.

The screenshot shows the Arduino IDE interface. The main editor window displays the file `micro_features_model.cpp`, which contains a large array of hex values representing a TFLite model. The array starts at line 1574 and ends at line 1601. The values are 8-bit integers represented in hexadecimal, such as `0x00, 0x00, 0x00, 0x01, 0x14, 0x00, 0x00, 0x00, 0x34, 0x00, 0x00, 0x00`.

Below the editor, the Serial Monitor is open, showing a series of messages received from the Arduino Nano 33 BLE. The messages are timestamped and include the text "Heard unknown", "Heard yes", and "Heard no". The baud rate is set to 115200.

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1574 0x00, 0x00, 0x00, 0x01, 0x14, 0x00, 0x00, 0x00, 0x34, 0x00, 0x00, 0x00,
1575 0x05, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x34, 0x00, 0x00, 0x00,
1576 0xbc, 0xfe, 0xff, 0xff, 0x00, 0x00, 0x00, 0x14, 0x00, 0x00, 0x00, 0x00,
1577 0x01, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00,
1578 0x00, 0x00, 0x00, 0x01, 0x00, 0x00, 0x00, 0xb5, 0xfa, 0xfa, 0x39,
1579 0x06, 0x00, 0x00, 0x00, 0x4d, 0x61, 0x74, 0x4d, 0x75, 0x6c, 0x00, 0x00,
1580 0x02, 0x00, 0x00, 0x00, 0x04, 0x00, 0x00, 0x00, 0xa0, 0x0f, 0x00, 0x00,
1581 0x2a, 0xff, 0xff, 0xff, 0x00, 0x00, 0x00, 0x01, 0x14, 0x00, 0x00, 0x00,
1582 0x30, 0x00, 0x00, 0x00, 0x04, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x02,
1583 0x38, 0x00, 0x00, 0x00, 0x14, 0xff, 0xff, 0xff, 0x00, 0x00, 0x00, 0x00,
1584 0x10, 0x00, 0x00, 0x00, 0x01, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00,
1585 0x00, 0x00, 0x00, 0x01, 0x00, 0x00, 0x00, 0x00, 0x10, 0x28, 0x38,
1586 0x0d, 0x00, 0x00, 0x00, 0x66, 0x69, 0x6e, 0x01, 0x6c, 0x5f, 0x66, 0x63,
1587 0x5f, 0x62, 0x69, 0x61, 0x72, 0x00, 0x00, 0x00, 0x01, 0x00, 0x00, 0x00,
1588 0x04, 0x00, 0x00, 0x00, 0x02, 0xff, 0xff, 0xff, 0x00, 0x00, 0x00, 0x01,
1589 0x14, 0x00, 0x00, 0x00, 0x14, 0x00, 0x00, 0x00, 0x03, 0x00, 0x00, 0x00,
1590 0x00, 0x00, 0x00, 0x02, 0x1c, 0x00, 0x00, 0x00, 0xc4, 0xff, 0xff, 0xff,
1591 0x0f, 0x00, 0x00, 0x00, 0x52, 0x65, 0x73, 0x68, 0x61, 0x70, 0x65, 0x5f,
1592 0x33, 0x2f, 0x73, 0x68, 0x61, 0x70, 0x65, 0x00, 0x01, 0x00, 0x00, 0x00,
1593 0x02, 0x00, 0x00, 0x00, 0xbe, 0xff, 0xff, 0xff, 0x00, 0x00, 0x00, 0x01,
1594 0x18, 0x00, 0x00, 0x00, 0x18, 0x00, 0x00, 0x00, 0x02, 0x00, 0x00, 0x00,
1595 0x00, 0x00, 0x00, 0x02, 0x20, 0x00, 0x00, 0x00, 0x04, 0x00, 0x04, 0x00,
1596 0x04, 0x00, 0x00, 0x00, 0x0f, 0x00, 0x00, 0x52, 0x65, 0x73, 0x68,
1597 0x61, 0x70, 0x65, 0x5f, 0x32, 0x2f, 0x73, 0x68, 0x61, 0x70, 0x65, 0x00,
1598 0x01, 0x00, 0x00, 0x00, 0x04, 0x00, 0x00, 0x00, 0x00, 0x00, 0x16, 0x00,
1599 0x1c, 0x00, 0x18, 0x00, 0x17, 0x00, 0x10, 0x00, 0xc, 0x00, 0x08, 0x00,
1600 0x00, 0x00, 0x00, 0x00, 0x00, 0x07, 0x00, 0x16, 0x00, 0x00, 0x00,
1601 0x00, 0x00, 0x00, 0x01, 0x20, 0x00, 0x00, 0x00, 0x40, 0x00, 0x00, 0x00,

```

Serial Monitor Output:

```

Message (Enter to send message to 'Arduino Nano 33 BLE' on '/dev/cu.usbmodem3101')
Heard unknown (201) @229840ms
Heard unknown (225) @231360ms
Heard unknown (201) @234048ms
Heard unknown (207) @236112ms
Heard unknown (201) @240960ms
Heard unknown (231) @242480ms
Heard unknown (201) @244272ms
Heard unknown (202) @246608ms
Heard unknown (201) @251632ms
Heard yes (209) @253968ms
Heard unknown (202) @257632ms
Heard unknown (208) @261040ms
Heard no (202) @263200ms
Heard unknown (209) @264720ms

```

6.